

Exam 1 - Part B

⚠ This is a preview of the draft version of the quiz

The Quiz is Part B (the second of three parts) of Exam 1.

Once you begin this quiz, you will have 20 minutes to complete it. The 20 minutes begins when you start the quiz, so read the instructions before you start. These instructions are the same for all parts of the exam.

Be sure to complete **all** parts of the exam before 9:00pm (Madison time).

Please remember that during the exam window, you are not permitted to communicate with other students in class. Note: if someone breaks this rule (e.g., by sending a message or posting

- Quiz Type

Graded Quiz
- Points

31
- Assignment Group

Exams
- Shuffle Answers

No
- Time Limit

20 Minutes
- Multiple Attempts

No
- View Responses

Always
- Show Correct Answers

Immediately
- One Question at a Time

No

Due	For	Available from	Until
Mar 8 at 9pm	Everyone	Mar 8 at 5:30pm	Mar 8 at 9pm

Preview

Score for this quiz: 0 out of 31

Submitted Mar 8 at 1:44pm

This attempt took less than 1 minute.

Unanswered

Question 1

0 / 2 pts

What color is the square that is farthest to the **right**? Note: squares might be drawn on top of one another.

We are asking about visible squares.

```
1 context.translate(200,200);
2 context.strokeStyle = "black";
3 for(let color of ["red", "green", "blue", "purple" ]) {
4   context.strokeRect(0,0, 10,10);
5   context.scale(-1,1);
6   context.translate(20,0);
7   context.strokeStyle = color;
8 }
```

☐ red

Correct Answer

☐ green

☐ blue

☐ purple

☐ black

This draws 4 squares... black, red, green, blue.

It alternates drawing them at x=200 (the starting point) and 180 (translate by $-1 * 20$).

The black and green squares are the right, the green covers the black, so the answer is green

Unanswered

Question 2

0 / 3 pts

Suppose you have a 2D rotation matrix (4 numbers) that rotated a certain amount. But, you would like to rotate by twice that amount. Which of the following could you do?

(check all that apply)

☐ Multiply the matrix by 2

Correct Answer

☐

Extract the angle from the matrix, multiply the angle by 2, and apply that rotation

Correct Answer

☐ Apply the rotation matrix twice

Unanswered

Question 3

0 / 4 pts

The 2D point 4,8 is transformed by the homogeneous transformation

matrix $\begin{bmatrix} 2 & 2 & 4 \\ 1 & 1 & 4 \\ 0 & 0 & 4 \end{bmatrix}$

The resulting point is

 ,

Answer 1:

Unanswered

(You left this blank)

Correct Answer

7

Answer 2:

Unanswered

(You left this blank)

Correct Answer

4

Unanswered

Question 4

0 / 5 pts

A cubic Bezier curve has its control points (0,0), (2,2), (4,4), (6,14).

If you cut the curve in half at $u=.5$, the first piece of the curve has its points at:

(0,0),

(,),

(,),

(,)

Answer 1:

You Answered

(You left this blank)

Correct Answer

1

Answer 2:

You Answered

(You left this blank)

Correct Answer

1

Answer 3:

You Answered

(You left this blank)

Correct Answer

2

Answer 4:

You Answered

(You left this blank)

Correct Answer

2

Answer 5:

You Answered

(You left this blank)

Correct Answer

3

Answer 6:

You Answered

(You left this blank)

Correct Answer

4

For X:

0 2 4 6

1 3 5

2 4

3

For Y:

0 2 4 14

1 3 9

2 6

4

So the points are (0,0), (1,1), (2,2), (3,4)

Unanswered

Question 5

0 / 6 pts

This program:

```

context.beginPath();
context.save();
context.scale(2,2);
context.translate(0,5);
context.rotate(Math.PI/2); // 90 degrees
context.translate(0,10);
context.scale(2,2);
( context.rotate(Math.PI/2); // 90 degrees
context.moveTo(-10,0);
context.restore();
context.rotate(-Math.PI/2); // -90 degrees
context.scale(-2,-2);
context.rotate(-Math.PI/2); // -90 degrees
context.lineTo(5,10);
context.stroke();

```

would be the same as:

```

context.beginPath();
context.moveTo( , );
context.lineTo( , );
context.stroke();

```

Answer 1:

ou Answered

(You left this blank)

orrect Answer

20

Answer 2:

ou Answered

(You left this blank)

orrect Answer

10

Answer 3:

You Answered

(You left this blank)

Correct Answer

10

Answer 4:

You Answered

(You left this blank)

Correct Answer

20

Unanswered

Question 6

0 / 5 pts

This program prints several lines to the console. The lines all have a letter (A or B) and a number.

```
1 function f(x) {  
2     let y = x;  
3     let g = function(z) {  
4         console.log("A",y+z);  
5         y = y+2;  
6     }  
7     y = y+1  
8     console.log("B",y);  
9     return g;  
10 }  
11 let a = f(10);  
12 let b = f(20);  
13 a(1);  
14 b(1);  
15 a(1);
```

Answer the following questions with numbers:

The letter A is printed times total.

The letter B is printed times total.

The letter A is printed times **before** the last time letter B is printed.

The number printed with the first A is .

The number printed with the last A is .

Answer 1:

You Answered

(You left this blank)

Correct Answer

3

Answer 2:

You Answered

(You left this blank)

Correct Answer

2

Answer 3:

You Answered

(You left this blank)

Correct Answer

0

Answer 4:

You Answered

(You left this blank)

Correct Answer

12

Answer 5:

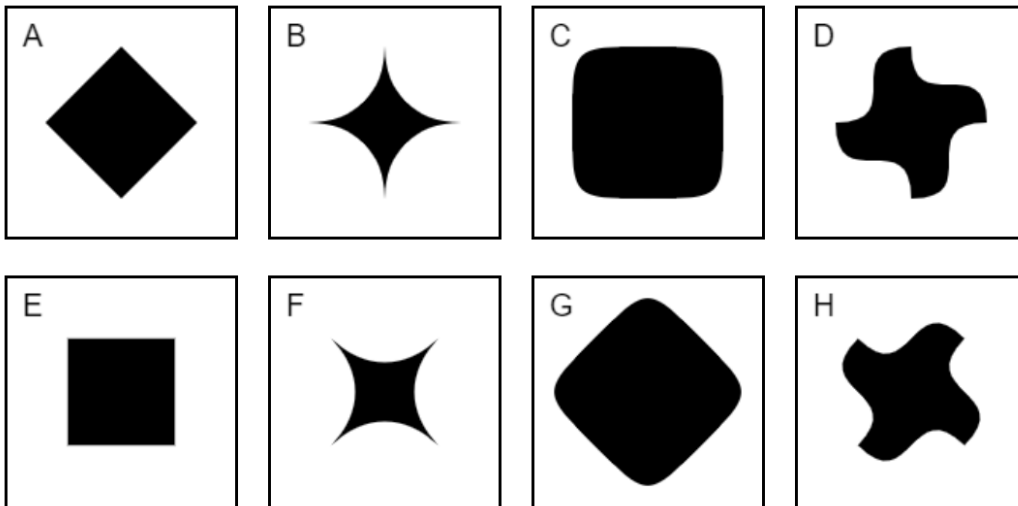
You Answered

(You left this blank)

Correct Answer

14

The next two questions refer to these 8 image choices (labeled A-H):



You Answered

Question 7

0 / 3 pts

Which shape (A-H) is drawn by:

```
context.translate(100,100);
context.moveTo(50,0);
for(let i=0; i<4; i++) {
    context.bezierCurveTo(50,0, 0,50, 0,50);
    context.rotate(Math.PI/2);
}
context.closePath();
context.fill();
```

Correct Answer

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

Unanswered

Question 8

0 / 3 pts

Which shape (A-H) is drawn by:

```
context.translate(100,100);
context.moveTo(50,0);
for(let i=0; i<4; i++) {
    context.bezierCurveTo(25,0, 0,25, 0,50);
    context.rotate(Math.PI/2); // 90 degrees
}
context.closePath();
context.fill();
```

☐ A

Correct Answer

☐ B☐ C☐ D

☐ E

☐ F

☐ G

☐ H

Quiz Score: **0** out of 31

